

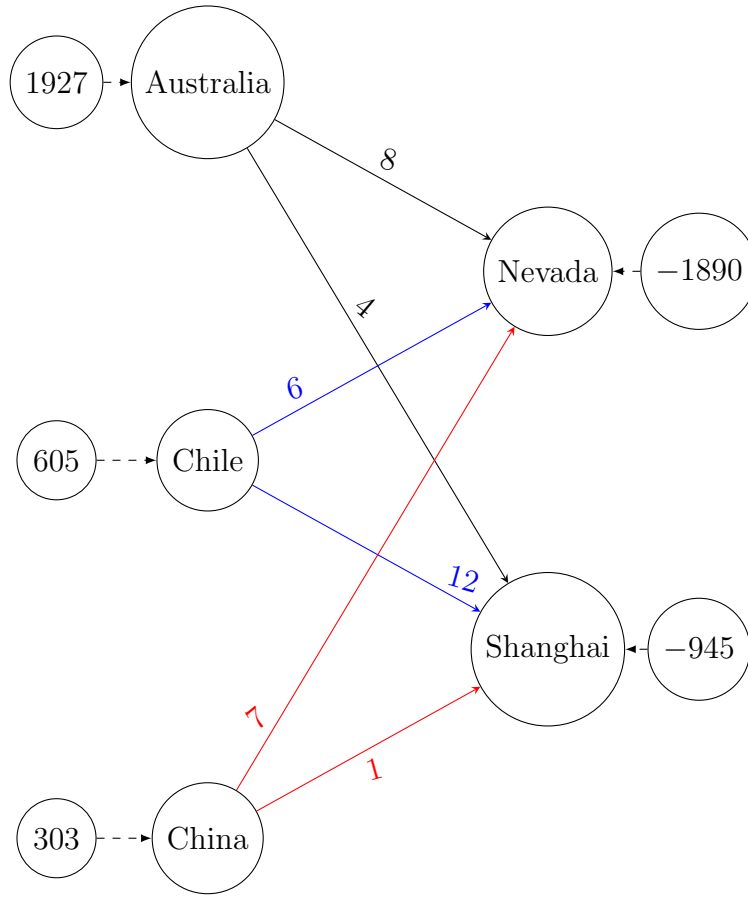
Modeling: transportation problem

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Solution of the practice quiz

The problem is not associated with a physical network. Still, we define a network in the following way. We define a node for each factory, represented on the right side of the network below, and a node for each mine, represented on the left side.

We introduce an arc between each pair of mine and factory. For example, the arc from Australia to Nevada means that the Australian mine is providing lithium to the Nevada factory. The cost associated with each arc is the transportation price per ton.



A transshipment problem is written

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}.$$

subject to the flow conservation constraints,

$$\text{div}(x)_i = s_i \quad \forall i \in \mathcal{N},$$

and the bounds on the flows:

$$\ell_{ij} \leq x_{ij} \leq u_{ij}, \quad \forall (i,j) \in \mathcal{A}.$$

In our case,

- $\mathcal{N}$ is the set of nodes: Australia (A), Chile (L), China (C), Nevada (N), Shanghai (S).

- $\mathcal{A}$ is the set of possible transactions, that is any pair mine, factory.
- Costs: $c_{ij}, \forall (i, j) \in \mathcal{A}$ corresponding to transportation prices per ton, as reported in the problem description.
- Supply: For each mine, the supply in tons is provided in the problem description: 1927 for Australia, 605 for Chile and 303 for China. The demand in lithium for Nevada is $30000 \cdot 0.063 = 1890$ tons. For Shanghai, it is $15000 \cdot 0.063 = 945$ tons of lithium. For each factory, the supply is the demand with the opposite sign, that is -1890 for Nevada, -945 for Shanghai.
- Bounds: As the transactions can only occur in one direction, we impose $x_{ij} \geq 0, \forall (i, j) \in \mathcal{A}$. No upper bound is necessary.

The transshipment problem is

$$\min_{x \in \mathbb{R}^6} 8x_{AN} + 4x_{AS} + 6x_{LN} + 12x_{LS} + 7x_{CN} + x_{CS},$$

subject to

$$\begin{aligned} x_{AN} + x_{AS} &= 1927, \\ x_{LN} + x_{LS} &= 605, \\ x_{CN} + x_{CS} &= 303, \\ -x_{AN} - x_{LN} - x_{CN} &= -1890, \\ -x_{AS} - x_{LS} - x_{CS} &= -945, \\ x_{ij} &\geq 0, \quad \forall (i, j) \in \mathcal{A}. \end{aligned}$$

The problem is in standard form. In order to apply the simplex algorithm, we need to modify it such that the right-hand side contains only non negative values by multiplying the corresponding constraints by -1 . If we do so, and number the variables in the following way:

$$\begin{aligned} x_1 &= x_{AN}, & x_2 &= x_{AS}, \\ x_3 &= x_{LN}, & x_4 &= x_{LS}, \\ x_5 &= x_{CN}, & x_6 &= x_{CS}, \end{aligned}$$

we obtain the following data:

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}, b = \begin{pmatrix} 1927 \\ 605 \\ 303 \\ 1890 \\ 945 \end{pmatrix}$$

and

$$c^T = (8 \quad 4 \quad 6 \quad 12 \quad 7 \quad 1).$$

An auxiliary variable is associated with each of the equality constraints $(x_7, \dots, x_{11})$ and the Phase I can be started.

Iteration 0

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	x_{11}	
1	1	0	0	0	0	1	0	0	0	0	1927
0	0	1	1	0	0	0	1	0	0	0	605
0	0	0	0	1	1	0	0	1	0	0	303
1	0	1	0	1	0	0	0	0	1	0	1890
0	1	0	1	0	1	0	0	0	0	1	945
-2	-2	-2	-2	-2	-2	0	0	0	0	0	-5670

Iteration 1

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	x_{11}	
0	1	-1	0	-1	0	1	0	0	-1	0	37
0	0	1	1	0	0	0	1	0	0	0	605
0	0	0	0	1	1	0	0	1	0	0	303
1	0	1	0	1	0	0	0	0	1	0	1890
0	1	0	1	0	1	0	0	0	0	1	945
0	-2	0	-2	0	-2	0	0	0	2	0	-1890

Iteration 2

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	x_{11}	
0	1	-1	0	-1	0	1	0	0	-1	0	37
0	0	1	1	0	0	0	1	0	0	0	605
0	0	0	0	1	1	0	0	1	0	0	303
1	0	1	0	1	0	0	0	0	1	0	1890
0	0	1	1	1	1	-1	0	0	1	1	908
0	0	-2	-2	-2	-2	2	0	0	0	0	-1816

Iteration 3

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	x_{11}	
0	1	0	1	-1	0	1	1	0	-1	0	642
0	0	1	1	0	0	0	1	0	0	0	605
0	0	0	0	1	1	0	0	1	0	0	303
1	0	0	-1	1	0	0	-1	0	1	0	1285
0	0	0	0	1	1	-1	-1	0	1	1	303
0	0	0	0	-2	-2	2	2	0	0	0	-606

Iteration 4

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}	x_{11}	
0	1	0	1	0	1	1	1	1	-1	0	945
0	0	1	1	0	0	0	1	0	0	0	605
0	0	0	0	1	1	0	0	1	0	0	303
1	0	0	-1	0	-1	0	-1	-1	1	0	982
0	0	0	0	0	0	-1	-1	-1	1	1	0
0	0	0	0	0	0	2	2	2	0	0	0

The optimal value is zero, and we have found a feasible solution of the original problem. The auxiliary x_{11} is in the basis, and cannot be pivoted out. It corresponds to the redundant constraint, that can be removed. After cleaning and calculation of the reduced costs, we can apply Phase II:

Phase II: Iteration 0

x_1	x_2	x_3	x_4	x_5	x_6	
0	1	0	1	0	1	945
0	0	1	1	0	0	605
0	0	0	0	1	1	303
1	0	0	-1	0	-1	982
0	0	0	10	0	-2	-17387

Phase II: Iteration 1

x_1	x_2	x_3	x_4	x_5	x_6	
0	1	0	1	-1	0	642
0	0	1	1	0	0	605
0	0	0	0	1	1	303
1	0	0	-1	1	0	1285
0	0	0	10	2	0	-16781

The tableau is optimal. Therefore, the total cost is \$16781, and the optimal strategy for the transportation of lithium is summarized in the following table:

	Lithium (tons)		
	Mines		
Factories	Australia	Chile	China
Nevada	1285	605	0
Shanghai	642	0	303